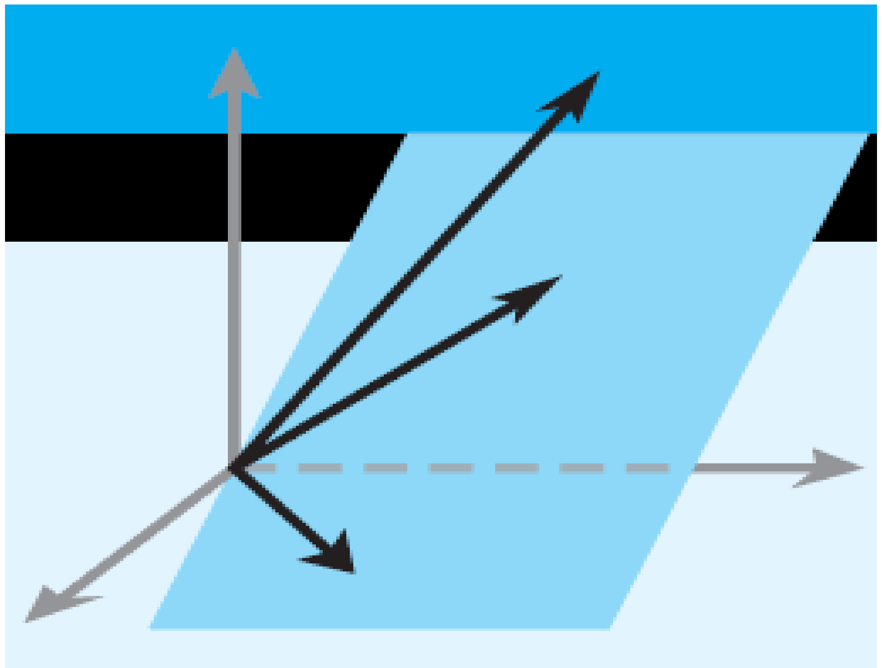
Elementary Linear Algebra



Chapter 6

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Chapter 6 Inner Product Spaces

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Gram-Schmidt Process; QR-Decomposition

DEFINITION 1

A set of two or more vectors in a real inner product space is said to be ***orthogonal*** if all pairs of distinct vectors in the set are orthogonal. An orthogonal set in which each vector has norm 1 is said to be ***orthonormal***.

EXAMPLE 1 **An Orthogonal Set in R^3**

Let

$$v_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad v_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad v_3 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

$$\mathbf{v}_1 = (0, 1, 0), \quad \mathbf{v}_2 = (1, 0, 1), \quad \mathbf{v}_3 = (1, 0, -1)$$

and assume that R^3 has the Euclidean inner product. It follows that the set of vectors $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is orthogonal since $\langle \mathbf{v}_1, \mathbf{v}_2 \rangle = \langle \mathbf{v}_1, \mathbf{v}_3 \rangle = \langle \mathbf{v}_2, \mathbf{v}_3 \rangle = 0$.

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EXAMPLE 2 Constructing an Orthonormal Set

The Euclidean norms of the vectors in Example 1 are

$$\| \mathbf{v}_1 \| = 1, \ \| \mathbf{v}_2 \| = \sqrt{2}, \ \| \mathbf{v}_3 \| = \sqrt{2}$$

Consequently, normalizing $\mathbf{u}_1$, $\mathbf{u}_2$, and $\mathbf{u}_3$ yields

$$\mathbf{u}_1 = \frac{\mathbf{v}_1}{\| \mathbf{v}_1 \|} = (0, 1, 0), \ \mathbf{u}_2 = \frac{\mathbf{v}_2}{\| \mathbf{v}_2 \|} = \left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}} \right),$$

$$\mathbf{u}_3 = \frac{\mathbf{v}_3}{\| \mathbf{v}_3 \|} = \left(\frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}} \right)$$

$$u_2 = \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ 1//\sqrt{2} \end{bmatrix}$$

$$u_3 = \begin{bmatrix} 1//\sqrt{2} \\ 0 \\ -1//\sqrt{2} \end{bmatrix}$$

We leave it for you to verify that the set $S = \{ \mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3 \}$ is orthonormal by showing that

$$\langle \mathbf{u}_1, \mathbf{u}_2 \rangle = \langle \mathbf{u}_1, \mathbf{u}_3 \rangle = \langle \mathbf{u}_2, \mathbf{u}_3 \rangle = 0 \quad \text{and} \quad \| \mathbf{u}_1 \| = \| \mathbf{u}_2 \| = \| \mathbf{u}_3 \| = 1$$

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THEOREM 6.3.1

If $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is an orthogonal set of nonzero vectors in an inner product space, then S is linearly independent.

Review

DEFINITION 1 If $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_r\}$ is a nonempty set of vectors in a vector space V, then the vector equation

$$k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + \cdots + k_r\mathbf{v}_r = \mathbf{0}$$

has at least one solution, namely,

$$k_1 = 0, \quad k_2 = 0, \ldots, \quad k_r = 0$$

We call this the ***trivial solution***. If this is the only solution, then S is said to be a ***linearly independent set***. If there are solutions in addition to the trivial solution, then S is said to be a ***linearly dependent set***.

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THEOREM 6.3.1

If $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is an orthogonal set of nonzero vectors in an inner product space, then S is linearly independent.

Proof: ***By definition we have to check***

$$k_1 \mathrm{v}_1 + k_2 \mathrm{v}_2 + \cdots + k_n \mathrm{v}_n = 0 \Rightarrow k_1 = 0, k_2 = 0, \cdots, k_n = 0$$

$$\Rightarrow < k_1 \mathrm{v}_1 + k_2 \mathrm{v}_2 + \cdots + k_n \mathrm{v}_n, \mathrm{v}_1 > = < 0, \mathrm{v}_1 >$$

$$\Rightarrow < k_1 \mathrm{v}_1 + k_2 \mathrm{v}_2 + \cdots + k_n \mathrm{v}_n, \mathrm{v}_1 > = 0$$ ***by Theorem 6.1.2***

$$\Rightarrow k_1 < \mathrm{v}_1, \mathrm{v}_1 > + k_2 < \mathrm{v}_2, \mathrm{v}_1 > + \cdots + k_n < \mathrm{v}_n, \mathrm{v}_1 > = 0$$

$$\Rightarrow k_1 < \mathrm{v}_1, \mathrm{v}_1 > + k_2 0 + \cdots + k_n 0 = 0$$ ***by orthogonal***

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THEOREM 6.3.1

If $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is an orthogonal set of nonzero vectors in an inner product space, then S is linearly independent.

Proof (conti):

$$k_1 < \mathrm{v}_1, \mathrm{v}_1 > + k_2 < \mathrm{v}_2, \mathrm{v}_1 > + \cdots + k_n < \mathrm{v}_n, \mathrm{v}_1 >= 0$$

$$\Rightarrow k_1 < \mathrm{v}_1, \mathrm{v}_1 > + k_2 0 + \cdots + k_n 0 = 0$$ ***by orthogonal***

$$\Rightarrow k_1 \| \mathrm{v}_1 \|^2 = 0$$

$$\Rightarrow k_1 = 0 \qquad \textit{because } \| \mathrm{v}_1 \|^2 \neq 0$$

Similarly, $k_1 = 0, k_2 = 0, \cdots, k_n = 0$

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EXAMPLE 3 An Orthonormal Basis

In Example 2 we showed that the vectors

$$\mathbf{u}_1 = (0, 1, 0), \quad \mathbf{u}_2 = \left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right), \quad \text{and} \quad \mathbf{u}_3 = \left(\frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}}\right)$$

form an orthonormal set with respect to the Euclidean inner product on R^3. By Theorem 6.3.1, these vectors form a linearly independent set, and since R^3 is three-dimensional, it follows from Theorem 4.5.4 that $S = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is an orthonormal basis for R^3.

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R^3

$$e_1 = \begin{bmatrix}1\\0\\0\end{bmatrix} \quad e_2 = \begin{bmatrix}0\\1\\0\end{bmatrix} \quad e_3 = \begin{bmatrix}0\\0\\1\end{bmatrix} \Rightarrow k_1\begin{bmatrix}1\\0\\0\end{bmatrix} + k_2\begin{bmatrix}0\\1\\0\end{bmatrix} + k_3\begin{bmatrix}0\\0\\1\end{bmatrix} \quad \mathrm{u} = \begin{bmatrix}1\\1\\1\end{bmatrix}$$

$$k_1 = ?, k_2 = ?, k_3 = ?$$

$$\mathrm{v}_1 = \begin{bmatrix}0\\1\\0\end{bmatrix} \quad \mathrm{v}_2 = \begin{bmatrix}1\\0\\1\end{bmatrix} \quad \mathrm{v}_3 = \begin{bmatrix}1\\0\\-1\end{bmatrix} \Rightarrow k_1\begin{bmatrix}0\\1\\0\end{bmatrix} + k_2\begin{bmatrix}1\\0\\1\end{bmatrix} + k_3\begin{bmatrix}1\\0\\-1\end{bmatrix} = \begin{bmatrix}1\\1\\1\end{bmatrix}$$

$$\Rightarrow k_1 = ?, k_2 = ?, k_3 = ? \Rightarrow \begin{bmatrix}0 & 1 & 1\\1 & 0 & 0\\0 & 1 & -1\end{bmatrix}\begin{bmatrix}k_1\\k_2\\k_3\end{bmatrix} = \begin{bmatrix}1\\1\\1\end{bmatrix}$$

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THEOREM 6.3.2

(a) If $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is an orthogonal basis for an inner product space V, and if $\mathbf{u}$ is any vector in V, then

$$\mathbf{u} = \frac{\langle \mathbf{u}, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2}\mathbf{v}_1 + \frac{\langle \mathbf{u}, \mathbf{v}_2 \rangle}{\|\mathbf{v}_2\|^2}\mathbf{v}_2 + \cdots + \frac{\langle \mathbf{u}, \mathbf{v}_n \rangle}{\|\mathbf{v}_n\|^2}\mathbf{v}_n \tag{2}$$

(b) If $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is an orthonormal basis for an inner product space V, and if $\mathbf{u}$ is any vector in V, then

$$\mathbf{u} = \langle \mathbf{u}, \mathbf{v}_1 \rangle \mathbf{v}_1 + \langle \mathbf{u}, \mathbf{v}_2 \rangle \mathbf{v}_2 + \cdots + \langle \mathbf{u}, \mathbf{v}_n \rangle \mathbf{v}_n \tag{3}$$

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Proof: ***By definition we have***

$$u = k_1 v_1 + k_2 v_2 + \cdots + k_n v_n$$

$$\Rightarrow \; < u, v_1 > = < k_1 v_1 + k_2 v_2 + \cdots + k_n v_n, v_1 >$$

$$\Rightarrow \; < u, v_1 > = k_1 < v_1, v_1 > + k_2 < v_2, v_1 > + \cdots + k_n < v_n, v_1 >$$

$$\Rightarrow \; < u, v_1 > = k_1 < v_1, v_1 >$$ ***by orthogonal***

$$\Rightarrow \; k_1 = \frac{< u, v_1 >}{< v_1, v_1 >} \;\Rightarrow\; k_1 = \frac{< u, v_1 >}{\| v_1 \|^2}$$

Similarly, $$k_i = \frac{< u, v_i >}{\| v_i \|^2}$$

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EXAMPLE 4 A Coordinate Vector Relative to an Orthonormal Basis

$$u = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Let

$$\mathbf{v}_1 = (0, 1, 0), \quad \mathbf{v}_2 = \left(-\frac{4}{5}, 0, \frac{3}{5}\right), \quad \mathbf{v}_3 = \left(\frac{3}{5}, 0, \frac{4}{5}\right)$$

It is easy to check that $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is an orthonormal basis for R^3 with the Euclidean inner product. Express the vector $\mathbf{u} = (1, 1, 1)$ as a linear combination of the vectors in S, and find the coordinate vector $(\mathbf{u})_S$.

Solution

We leave it for you to verify that

$$\langle \mathbf{u}, \mathbf{v}_1 \rangle = 1, \quad \langle \mathbf{u}, \mathbf{v}_2 \rangle = -\frac{1}{5}, \quad \text{and} \quad \langle \mathbf{u}, \mathbf{v}_3 \rangle = \frac{7}{5}$$

Therefore, by Theorem 6.3.2 we have

$$\mathbf{u} = \mathbf{v}_1 - \frac{1}{5}\mathbf{v}_2 + \frac{7}{5}\mathbf{v}_3$$

$$v_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad v_2 = \begin{bmatrix} -4/5 \\ 0 \\ 3/5 \end{bmatrix} \quad v_3 = \begin{bmatrix} 3/5 \\ 0 \\ 4/5 \end{bmatrix}$$

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that is,

$$(1, 1, 1) = (0, 1, 0) - \frac{1}{5}\left(-\frac{4}{5}, 0, \frac{3}{5}\right) + \frac{7}{5}\left(\frac{3}{5}, 0, \frac{4}{5}\right)$$

Thus, the coordinate vector of $\mathbf{u}$ relative to S is

$$(\mathbf{u})_S = (\langle \mathbf{u}, \mathbf{v}_1 \rangle, \langle \mathbf{u}, \mathbf{v}_2 \rangle, \langle \mathbf{u}, \mathbf{v}_3 \rangle) = \left(1, -\frac{1}{5}, \frac{7}{5}\right)$$

$$v_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad v_2 = \begin{bmatrix} -4/5 \\ 0 \\ 3/5 \end{bmatrix} \quad v_3 = \begin{bmatrix} 3/5 \\ 0 \\ 4/5 \end{bmatrix}$$

$$u = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

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EXAMPLE 5 An Orthonormal Basis from an Orthogonal Basis

(a) Show that the vectors

$$\mathbf{w}_1 = (0, 2, 0), \quad \mathbf{w}_2 = (3, 0, 3), \quad \mathbf{w}_3 = (-4, 0, 4)$$

form an orthogonal basis for R^3 with the Euclidean inner product, and use that basis to find an orthonormal basis by normalizing each vector.

(b) Express the vector $\mathbf{u} = (1, 2, 4)$ as a linear combination of the orthonormal basis vectors obtained in part (a).

Solution (a) The given vectors form an orthogonal set since

$$\langle \mathbf{w}_1, \mathbf{w}_2 \rangle = 0, \quad \langle \mathbf{w}_1, \mathbf{w}_3 \rangle = 0, \quad \langle \mathbf{w}_2, \mathbf{w}_3 \rangle = 0$$

It follows from Theorem 6.3.1 that these vectors are linearly independent and hence form a basis for R^3 by Theorem 4.5.4. We leave it for you to calculate the norms of $\mathbf{w}_1$, $\mathbf{w}_2$, and $\mathbf{w}_3$ and then obtain the orthonormal basis

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►EXAMPLE 5

$$\mathbf{v}_1 = \frac{\mathbf{w}_1}{\|\mathbf{w}_1\|} = (0, 1, 0), \mathbf{v}_2 = \frac{\mathbf{w}_2}{\|\mathbf{w}_2\|} = \left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right),$$

$$\mathbf{v}_3 = \frac{\mathbf{w}_3}{\|\mathbf{w}_3\|} = \left(-\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right)$$

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EXAMPLE 5

Solution (b) It follows from Formula 4 that

$$\mathbf{u} = \langle \mathbf{u}, \mathbf{v}_1 \rangle \mathbf{v}_1 + \langle \mathbf{u}, \mathbf{v}_2 \rangle \mathbf{v}_2 + \langle \mathbf{u}, \mathbf{v}_3 \rangle \mathbf{v}_3$$

We leave it for you to confirm that

$$\begin{aligned}
\langle \mathbf{u}, \mathbf{v}_1 \rangle &= (1, 2, 4) \cdot (0, 1, 0) = 2 \\
\langle \mathbf{u}, \mathbf{v}_2 \rangle &= (1, 2, 4) \cdot \left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right) = \frac{5}{\sqrt{2}} \\
\langle \mathbf{u}, \mathbf{v}_3 \rangle &= (1, 2, 4) \cdot \left(-\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right) = \frac{3}{\sqrt{2}}
\end{aligned}$$

and hence that

$$(1, 2, 4) = 2(0, 1, 0) + \frac{5}{\sqrt{2}}\left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right) + \frac{3}{\sqrt{2}}\left(-\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right)$$

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THEOREM 6.3.3 Projection Theorem

If W is a finite-dimensional subspace of an inner product space V, then every vector $\mathbf{u}$ in V can be expressed in exactly one way as

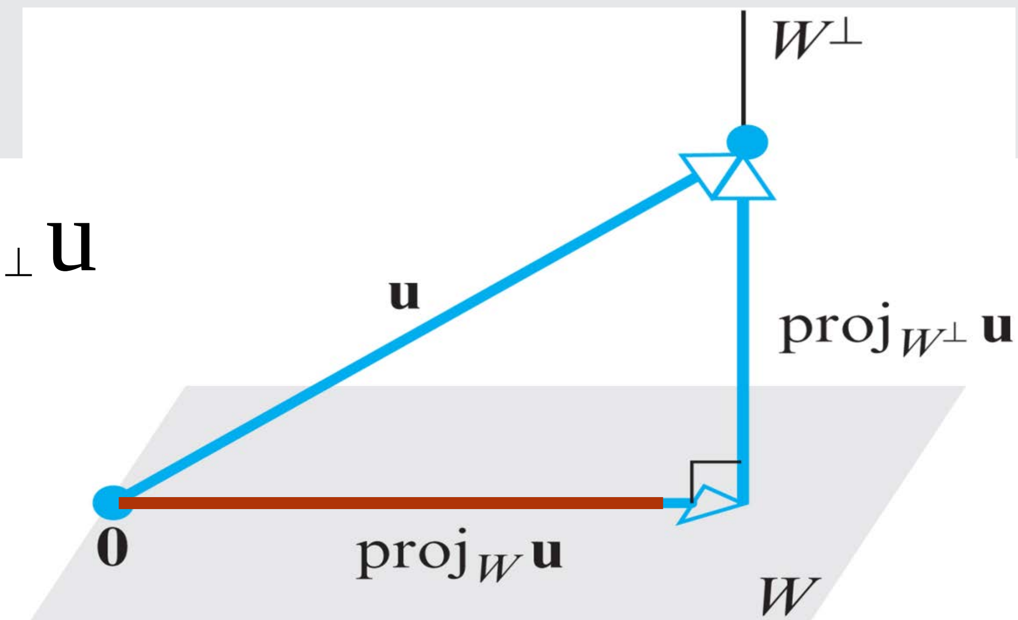
$$\mathbf{u} = \mathbf{w}_1 + \mathbf{w}_2 \tag{3}$$

where $\mathbf{w}_1$ is in W and $\mathbf{w}_2$ is in $W^{\perp}$.

$$\mathrm{w}_1 = \mathrm{proj}_W\, \mathrm{u} \qquad \mathrm{w}_2 = \mathrm{proj}_{W^{\perp}}\, \mathrm{u}$$

$$\mathrm{u} = \mathrm{proj}_W\, \mathrm{u} + \mathrm{proj}_{W^{\perp}}\, \mathrm{u}$$

$$\mathrm{proj}_{W^{\perp}}\, \mathrm{u} = \mathrm{u} - \mathrm{proj}_W\, \mathrm{u}$$



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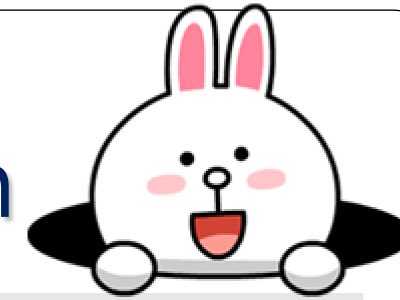

THEOREM 6.3.4

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Let W be a finite-dimensional subspace of an inner product space V.

(a) If $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_r\}$ is an orthogonal basis for W, and $\mathbf{u}$ is any vector in V, then

$$\text{proj}_W \mathbf{u} = \frac{\langle \mathbf{u}, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2} v_1 + \frac{\langle \mathbf{u}, \mathbf{v}_2 \rangle}{\|\mathbf{v}_2\|^2} \mathbf{v}_2 + \cdots + \frac{\langle \mathbf{u}, \mathbf{v}_r \rangle}{\|\mathbf{v}_r\|^2} v_r \qquad \textbf{(11)}$$

(b) If $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_r\}$ is an orthonormal basis for W, and $\mathbf{u}$ is any vector in V, then

$$\text{proj}_W \mathbf{u} = \langle \mathbf{u}, \mathbf{v}_1 \rangle \mathbf{v}_1 + \langle \mathbf{u}, \mathbf{v}_2 \rangle \mathbf{v}_2 + \cdots + \langle \mathbf{u}, \mathbf{v}_r \rangle \mathbf{v}_r \qquad \textbf{(12)}$$

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Proof: $\mathrm{w}_1 = \mathrm{proj}_W \mathrm{u} \quad \in W$

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$$= k_1 \mathrm{v}_1 + k_2 \mathrm{v}_2 + \cdots + k_r \mathrm{v}_r$$

$$= \frac{<\mathrm{w}_1, \mathrm{v}_1>}{\|\mathrm{v}_1\|^2}\mathrm{v}_1 + \frac{<\mathrm{w}_1, \mathrm{v}_2>}{\|\mathrm{v}_2\|^2}\mathrm{v}_2 + \quad \cdots \qquad + \frac{<\mathrm{w}_1, \mathrm{v}_r>}{\|\mathrm{v}_r\|^2}\mathrm{v}_r$$

by Theorem 6.3.2

$$\mathrm{w}_2 = \mathrm{proj}_{W^\perp} \mathrm{u} \quad \in W^\perp$$

$$\mathrm{v}_1 \in W, \mathrm{v}_2 \in W, \cdots \mathrm{v}_r \in W$$

$$<\mathrm{w}_2, \mathrm{v}_1>= \qquad <\mathrm{w}_2, \mathrm{v}_2>= \qquad <\mathrm{w}_n, \mathrm{v}_n>=$$

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$$w_1 = \mathrm{proj}_W\, u = \frac{<w_1, v_1>}{\| v_1 \|^2} v_1 + \frac{<w_1, v_2>}{\| v_2 \|^2} v_2 + \text{---------} + \frac{<w_1, v_r>}{\| v_r \|^2} v_r$$

$$= \frac{<w_1, v_1> + 0}{\| v_1 \|^2} v_1 + \frac{<w_1, v_2> + 0}{\| v_2 \|^2} v_2 + \text{---------} + \frac{<w_1, v_r> + 0}{\| v_r \|^2} v_r$$

$$= \frac{<w_1, v_1> + <w_2, v_1>}{\| v_1 \|^2} v_1 + \text{-------------} + \frac{<w_1, v_r> + <w_2, v_r>}{\| v_r \|^2} v_r$$

$$= \frac{<w_1 + w_2, v_1>}{\| v_1 \|^2} v_1 + \text{-------------} + \frac{<w_1 + w_2, v_r>}{\| v_r \|^2} v_r$$

$$= \frac{<u, v_1>}{\| v_1 \|^2} v_1 + \text{-------------} + \frac{<u, v_r>}{\| v_r \|^2} v_r$$

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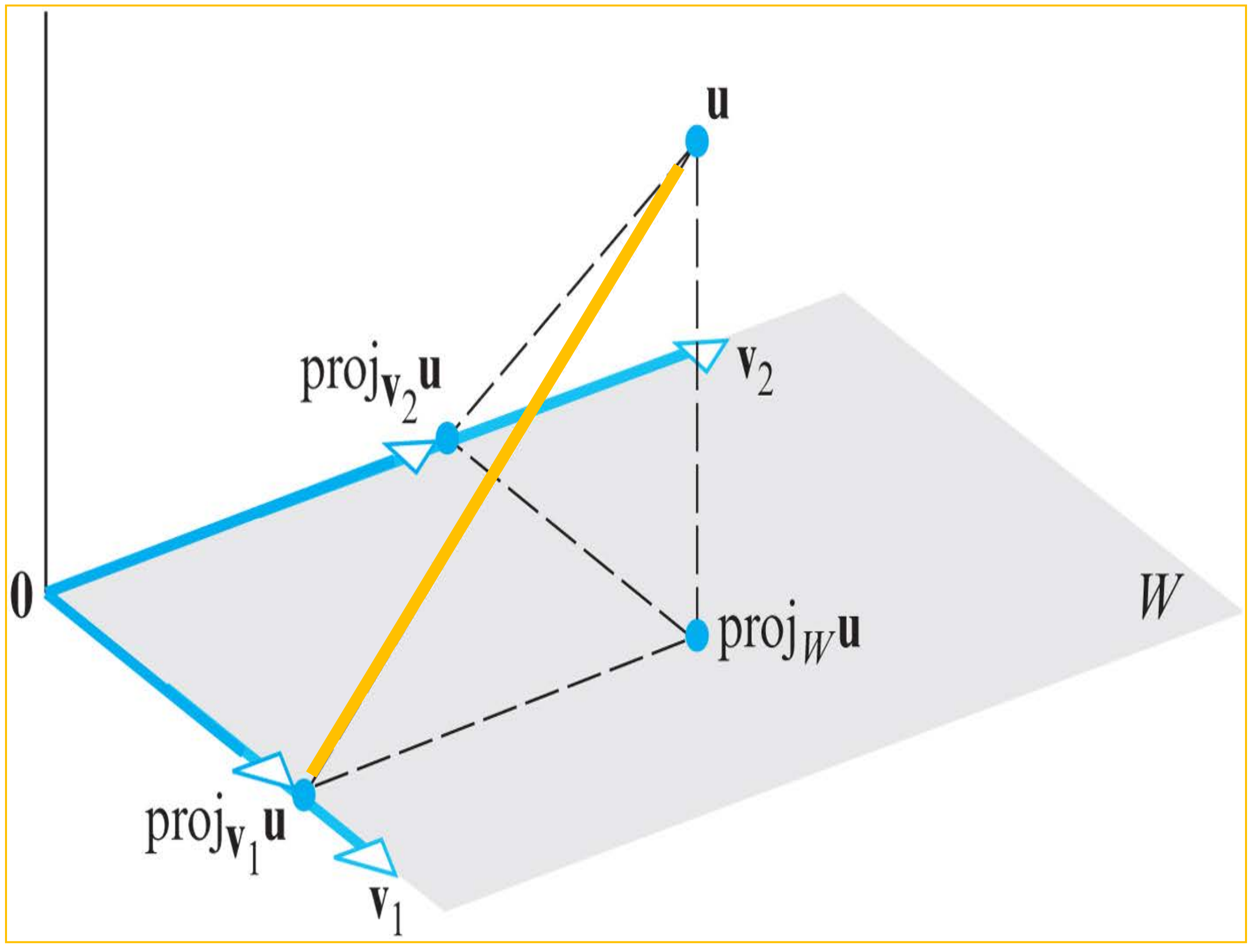
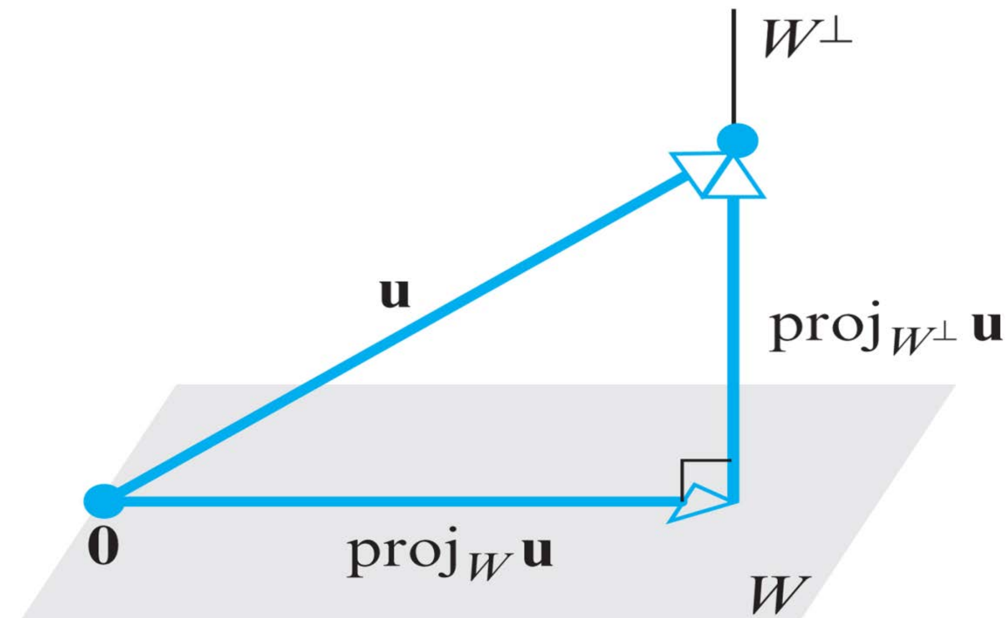
Geometric Interpretation of Orthogonal Projections





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EXAMPLE 6 Calculating Projections

Let R^3 have the Euclidean inner product, and let W be the subspace spanned by the orthonormal vectors $\mathbf{v}_1 = (0, 1, 0)$ and $\mathbf{v}_2 = (-4/5, 0, 3/5)$. From Formula (12) the orthog-onal projection of $\mathbf{u} = (1, 1, 1)$ on W is

$$\begin{aligned}
\operatorname{proj}_W \mathbf{u} &= \langle \mathbf{u}, \mathbf{v}_1 \rangle \mathbf{v}_1 + \langle \mathbf{u}, \mathbf{v}_2 \rangle \mathbf{v}_2 \\
&= (1)\,(0, 1, 0) + \left(-\frac{1}{5}\right)\left(-\frac{4}{5}, 0, \frac{3}{5}\right) \\
&= \left(\frac{4}{25}, 1, -\frac{3}{25}\right)
\end{aligned}$$

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The component of $\mathbf{u}$ orthogonal to W is

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$$\text{proj}_{W^\perp}\mathbf{u} = \mathbf{u} - \text{proj}_W\mathbf{u} = (1, 1, 1) - \left(\frac{4}{25}, 1, -\frac{3}{25}\right) = \left(\frac{21}{25}, 0, \frac{28}{25}\right)$$

Observe that $\text{pro}_{W\perp}\mathbf{u}$ is orthogonal to both $\mathbf{v}_1$ and $\mathbf{v}_2$, so this vector is orthogonal to each vector in the space W spanned by $\mathbf{v}_1$ and $\mathbf{v}_2$, as it should be.

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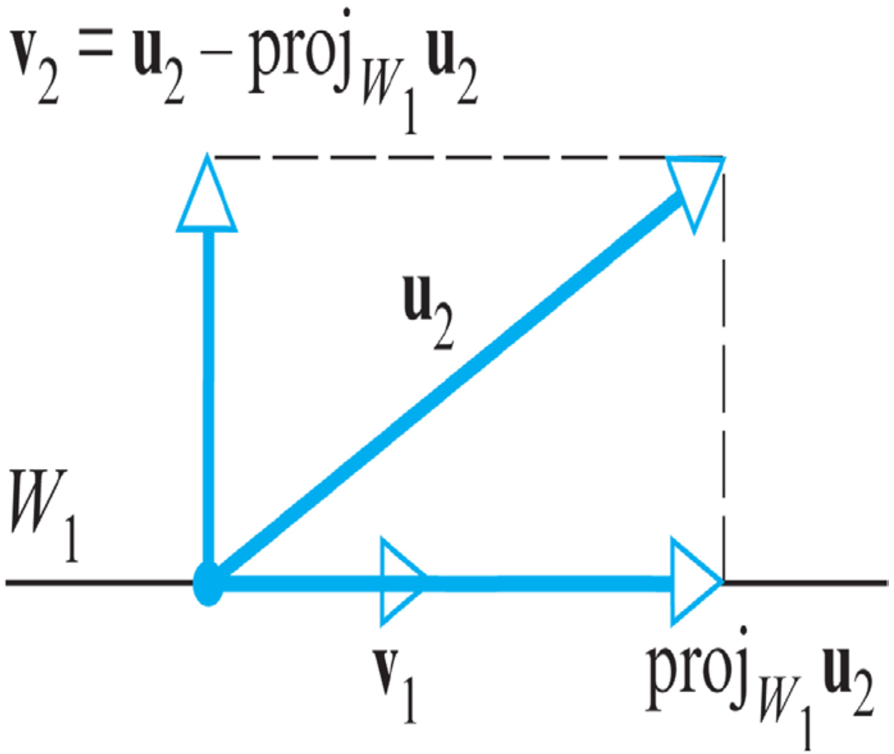
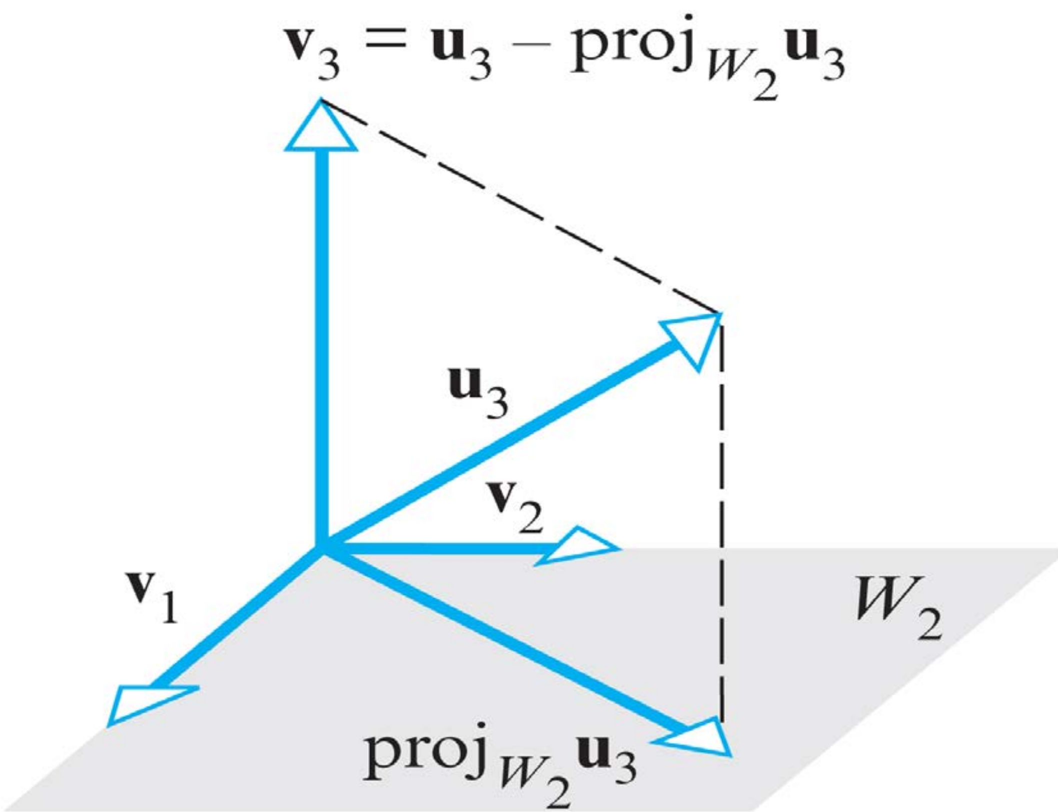
THEOREM 6.3.5 **Page 353**

Every nonzero finite-dimensional inner product space has an orthonormal basis.





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The Gram–Schmidt Process **Page 354**

To convert a basis $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_r\}$ into an orthogonal basis $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_r\}$, perform the following computations:

Step 1. $\mathbf{v}_1 = \mathbf{u}_1$

Step 2. $$\mathbf{v}_2 = \mathbf{u}_2 - \frac{\langle \mathbf{u}_2, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2}\mathbf{v}_1$$

Step 3. $$\mathbf{v}_3 = \mathbf{u}_3 - \frac{\langle \mathbf{u}_3, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2}\mathbf{v}_1 - \frac{\langle \mathbf{u}_3, \mathbf{v}_2 \rangle}{\|\mathbf{v}_2\|^2}\mathbf{v}_2$$

Step 4. $$\mathbf{v}_4 = \mathbf{u}_4 - \frac{\langle \mathbf{u}_4, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2}\mathbf{v}_1 - \frac{\langle \mathbf{u}_4, \mathbf{v}_2 \rangle}{\|\mathbf{v}_2\|^2}\mathbf{v}_2 - \frac{\langle \mathbf{u}_4, \mathbf{v}_3 \rangle}{\|\mathbf{v}_3\|^2}\mathbf{v}_3$$

$\vdots$

(continue for r steps)

Optional Step. To convert the orthogonal basis into an orthonormal basis $\{\mathbf{q}_1, \mathbf{q}_2, \ldots, \mathbf{q}_r\}$, normalize the orthogonal basis vectors.

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EXAMPLE 7 **Using the Gram-Schmidt Process**

Assume that the vector space R^3 has the Euclidean inner product. Apply the Gram-Schmidt process to transform the basis vectors

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$$\mathbf{u}_1 = (1, 1, 1), \quad \mathbf{u}_2 = (0, 1, 1), \quad \mathbf{u}_3 = (0, 0, 1)$$

into an orthogonal basis $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$, and then normalize the orthogonal basis vectors to obtain an orthonormal basis $\{\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3\}$.

$$u_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad u_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \quad u_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Solution

Step 1.

$$v_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \qquad \mathbf{v}_1 = \mathbf{u}_1 = (1, 1, 1)$$

Step 2.

$$\mathbf{v}_2 = \mathbf{u}_2 - \mathrm{proj}_{W_1}\mathbf{u}_2 = \mathbf{u}_2 - \frac{\langle \mathbf{u}_2, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2}\mathbf{v}_1 \qquad u_2 - \mathrm{proj}_{v_1} u_2$$

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▶ **EXAMPLE 7**

$$\boxed{\mathbf{v}_2 =} \ (0, 1, 1) - \frac{2}{3}(1, 1, 1) = \left(-\frac{2}{3}, \frac{1}{3}, \frac{1}{3}\right)$$

Step 3. $\boxed{\mathbf{v}_3 = \mathbf{u}_3 - \text{proj}_{\mathbf{v}_1}\mathbf{u}_3 - \text{proj}_{\mathbf{v}_2}\mathbf{u}_3}$

$$\begin{aligned}
\mathbf{v}_3 &= \mathbf{u}_3 - \text{proj}_{W_2}\mathbf{u}_3 = \mathbf{u}_3 - \frac{\langle \mathbf{u}_3, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2}\mathbf{v}_1 - \frac{\langle \mathbf{u}_3, \mathbf{v}_2 \rangle}{\|\mathbf{v}_2\|^2}\mathbf{v}_2 \\
&= (0, 0, 1) - \frac{1}{3}(1, 1, 1) - \frac{1/3}{2/3}\left(-\frac{2}{3}, \frac{1}{3}, \frac{1}{3}\right) \\
&= \left(0, -\frac{1}{2}, \frac{1}{2}\right)
\end{aligned}$$

Thus,

$$\mathbf{v}_1 = (1, 1, 1), \quad \mathbf{v}_2 = \left(-\frac{2}{3}, \frac{1}{3}, \frac{1}{3}\right), \quad \mathbf{v}_3 = \left(0, -\frac{1}{2}, \frac{1}{2}\right)$$

form an orthogonal basis for R^3. The norms of these vectors are

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EXAMPLE 7

$$\|\mathbf{v}_1\| = \sqrt{3}, \quad \|\mathbf{v}_2\| = \frac{\sqrt{6}}{3}, \quad \|\mathbf{v}_3\| = \frac{1}{\sqrt{2}}$$

so an orthonormal basis for R^3 is

$$\mathbf{q}_1 = \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|} = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right), \quad \mathbf{q}_2 = \frac{\mathbf{v}_2}{\|\mathbf{v}_2\|} = \left(-\frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right),$$

$$\mathbf{q}_3 = \frac{\mathbf{v}_3}{\|\mathbf{v}_3\|} = \left(0, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$$

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EXAMPLE 8 Legendre Polynomials

Let the vector space P_2 have the inner product

$$\langle \mathbf{p}, \mathbf{q} \rangle = \int_{-1}^{1} p(x)\, q(x)\, dx$$

Apply the Gram-Schmidt process to transform the standard basis $\{1, x, x^2\}$ for P_2 into an orthogonal basis $\{\phi_1(x), \phi_2(x), \phi_3(x)\}$.

Solution

Take $\mathbf{u}_1 = 1$, $\mathbf{u}_2 = x$, and $\mathbf{u}_3 = x^2$.

Step 1

$$\mathbf{v}_1 = \mathbf{u}_1 = 1$$

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Step 2

$$\langle \mathbf{u}_2, \mathbf{v}_1 \rangle = \int_{-1}^{1} x\,dx = 0$$

so

$$\mathbf{v}_2 = \mathbf{u}_2 - \frac{\langle \mathbf{u}_2, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2}\mathbf{v}_1 = \mathbf{u}_2 = x$$

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Step 3 We have

$$\langle \mathbf{u}_3, \mathbf{v}_1 \rangle = \int_{-1}^{1} x^2 dx = \frac{x^3}{3}\Big]_{-1}^{1} = \frac{2}{3}$$

$$\langle \mathbf{u}_3, \mathbf{v}_2 \rangle = \int_{-1}^{1} x^3 dx = \frac{x^4}{4}\Big]_{-1}^{1} = 0$$

$$\|\mathbf{v}_1\|^2 = \langle \mathbf{v}_1, \mathbf{v}_1 \rangle = \int_{-1}^{1} 1 dx = x\Big]_{-1}^{1} = 2$$

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so

$$\mathbf{v}_3 = \mathbf{u}_3 - \frac{\langle \mathbf{u}_3, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2}\mathbf{v}_1 - \frac{\langle \mathbf{u}_3, \mathbf{v}_2 \rangle}{\|\mathbf{v}_2\|^2}\mathbf{v}_2 = x^2 - \frac{1}{3}$$

Thus, we have obtained the orthogonal basis $\{\phi_1(x), \phi_2(x), \phi_3(x)\}$ in which

$$\phi_1(x) = 1, \quad \phi_2(x) = x, \quad \phi_3(x) = x^2 - \frac{1}{3}$$

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THEOREM 6.3.6

If W is a finite-dimensional inner product space, then:

(a) Every orthogonal set of nonzero vectors in W can be enlarged to an orthogonal basis for W.

(b) Every orthonormal set in W can be enlarged to an orthonormal basis for W.

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THEOREM 6.3.7 *QR*-Decomposition

If A is an $m \times n$ matrix with linearly independent column vectors, then A can be factored as

$$A = QR$$

where Q is an $m \times n$ matrix with orthonormal column vectors, and R is an $n \times n$ invertible upper triangular matrix.

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EXAMPLE 9 ***QR*-Decomposition of a 3 × 3 Matrix**

Find a *QR*-decomposition of

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

Solution The column vectors of A are

$$\mathbf{u}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{u}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{u}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

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Applying the Gram-Schmidt process with normalization to these column vectors yields the orthonormal vectors (see Example 7)

$$\mathbf{q}_1 = \begin{bmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{bmatrix}, \quad \mathbf{q}_2 = \begin{bmatrix} -\frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \end{bmatrix}, \quad \mathbf{q}_3 = \begin{bmatrix} 0 \\ -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

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Thus, it follows from Formula (15) that $\boldsymbol{R}$ is

$$R = \begin{bmatrix} \langle \mathbf{u}_1, \mathbf{q}_1 \rangle & \langle \mathbf{u}_2, \mathbf{q}_1 \rangle & \langle \mathbf{u}_3, \mathbf{q}_1 \rangle \\ 0 & \langle \mathbf{u}_2, \mathbf{q}_2 \rangle & \langle \mathbf{u}_3, \mathbf{q}_2 \rangle \\ 0 & 0 & \langle \mathbf{u}_3, \mathbf{q}_3 \rangle \end{bmatrix} = \begin{bmatrix} \frac{3}{\sqrt{3}} & \frac{2}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ 0 & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}$$

from which it follows that a QR-decomposition of $\boldsymbol{A}$ is

$$\underset{A}{\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}} = \underset{Q}{\begin{bmatrix} \frac{1}{\sqrt{3}} & -\frac{2}{\sqrt{6}} & 0 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} \end{bmatrix}} \underset{R}{\begin{bmatrix} \frac{3}{\sqrt{3}} & \frac{2}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ 0 & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}}$$

Homework

Sec 6.3

Practice for Quiz on 3/26 and Midterm on 4/18

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 # 80, 83, 85 Page 382

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